THE INERTIA CHARACTER IS EQUIVARIANT FOR THE ACTION OF THE DECOMPOSITION GROUP

THE STACKS PROJECT AUTHORS

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Editorial note. The statement and proof use the notation of the surrounding section for the decomposition group, the inertia group, the uniformizer and the residue field; those notations are introduced by the two lemmas the proof refers to, which are carried as navigation stubs.

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Allowed standard prerequisites (editorial).

- The inertia group is normal in the decomposition group, and the decomposition group acts on the residue field by the map to its automorphism group. used at proof lines 33442-33446. Explains why the formula of the statement is meaningful and what the action on the right hand side is.
- An inertia element acts on a uniformizer by multiplication by a unit, whose image in the residue field is the value of the inertia character. used at proof lines 33447-33453. Recalls the construction of the character, which the computation uses.
- The relation between the unit attached to an element of the decomposition group and the unit attached to its inverse, together with the triviality of inertia elements modulo the uniformizer. used at proof lines 33454-33468. Supplies the substitutions that turn the value at the conjugate into the conjugate of the value at the original element.

Source reference. Original lemma 09EE: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33251–33251. The reference is the construction of the inertia character recalled at proof lines 33447-33453; the computation of the conjugate that proves the statement is inside the counted span.

Source reference. Original lemma 09ED: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33224–33224. The reference supplies the action of the decomposition group on the residue field invoked at proof lines 33444-33446; it is a

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pointer needed to make the formula meaningful and the rest of the argument is inside the counted span.

Lemma 0.2. *With assumptions and notation as in Lemma 09EE. The inertia character $\theta : I \to \mu_e(\kappa(\mathfrak{m}))$ satisfies the following property*

$$\theta(\tau\sigma\tau^{-1}) = \tau(\theta(\sigma))$$

for $\tau \in D$ and $\sigma \in I$.

Proof. The formula makes sense as I is a normal subgroup of D and as τ acts on $\kappa(\mathfrak{m})$ via the map $D \to \mathrm{Aut}(\kappa(\mathfrak{m}))$ discussed in Lemma 09ED for example. Recall the construction of θ. Choose a uniformizer π of $B_{\mathfrak{m}}$ and for $\sigma \in I$ write $\sigma(\pi) = \theta_\sigma \pi$. Then $\theta(\sigma)$ is the image $\overline{\theta}_\sigma$ of θ_σ in the residue field. For any $\tau \in D$ we can write $\tau(\pi) = \theta_\tau \pi$ for some unit θ_τ. Then $\theta_{\tau^{-1}} = \tau^{-1}(\theta_\tau^{-1})$. We compute

$$\begin{aligned}
\theta_{\tau\sigma\tau^{-1}} &= \tau(\sigma(\tau^{-1}(\pi)))/\pi \\
&= \tau(\sigma(\tau^{-1}(\theta_\tau^{-1})\pi))/\pi \\
&= \tau(\sigma(\tau^{-1}(\theta_\tau^{-1}))\theta_\sigma\pi)/\pi \\
&= \tau(\sigma(\tau^{-1}(\theta_\tau^{-1})))\tau(\theta_\sigma)\theta_\tau
\end{aligned}$$

However, since σ acts trivially modulo π we see that the product $\tau(\sigma(\tau^{-1}(\theta_\tau^{-1})))\theta_\tau$ maps to 1 in the residue field. This proves the lemma. □